THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** time period	Time period is measured.
2	**(a)** 0.005 mm	2 / 400 = 0.005 mm.
3	**(c)** rash seen by the doctor	Signs are observed.
4	**(d)** break in the circuit	Loose contact stops current.
5	**(b)** 4 N	Balanced forces.
6	**(a)** 600 Pa	240 / 0.4 = 600.
7	**(c)** particles move faster at higher temperature	Higher kinetic energy.
8	**(b)** O_2 and Fe	Oxygen and iron.
9	**(d)** 20 per cent	40/200 x 100.
10	**(c)** 65 degrees	i = 90 - 25 = 65.
11	**(d)** 45 degrees	Shadow = height.
12	**(b)** small fish	Third level.
13	**(b)** less than 1 per cent of the radius	35/6400 is about 0.5 per cent.
14	**(d)** 20 degrees C per minute	Rise of 20 each minute.
15	**(d)** has similar cells doing one job	Organs have more than one tissue type.
16	**(c)** entry, multiplication, symptoms	Pathogen enters and multiplies before symptoms.
17	**(b)** heating	All use resistive heating.
18	**(d)** positive charge	Like charges repel.
19	**(d)** more air is pumped in	More collisions.
20	**(a)** gives out extra heat when it condenses on the skin	Condensation of steam releases the heat absorbed during vaporisation.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Independent variable is changed by us (1); Dependent variable changes as a result and is measured (1).	1+1
22	They lack a nucleus enclosed by a membrane (1); They are much simpler and smaller (1).	1+1
23	Mosquitoes breed in it (1); Malaria and dengue (1).	1+1
24	Fused filament or loose connection (1); Check connections and try another bulb (1).	1+1
25	Net = 15 - 6 = 9 N (1); In the direction of push (1).	1+1
26	P = 80/0.2 (1); = 400 Pa (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Change: temperature of water (1); Measure: time taken to dissolve (1); Same: mass of sugar, volume of water, stirring (1).	1+1+1
28	Plant cell (1); Chloroplasts (1); Make food by photosynthesis (1).	1+1+1
29	Vaccine contains weak pathogen or antigen (1); Body makes antibodies and memory cells (1); Boosters keep the memory strong (1).	1+1+1
30	Broken connection or weak cell: replace it (1); No iron core: use an iron nail (1); Too few turns: add turns (1).	1+1+1
31	25 N is static friction at the limit (1); 18 N is sliding friction (1); Static friction is larger than sliding friction (1).	1+1+1
	OR: P = 400 / 0.02 = 20000 Pa (1); Two feet: area 0.04, P = 10000 Pa (1); Pressure is double on one foot (1).	1+1+1
32	Solids: strong forces hold particles (1); Liquids: particles slide over each other (1); Gases: particles fly apart (1).	1+1+1
	OR: Black mass forms (1); Magnet does not attract it (1); A compound is formed (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Measure temperature fall with a thermometer (1); Constant: volume, vessel, starting temperature, place (1); Heat each to same temperature (1); Record every minute in a table (1); Draw a graph and conclude (1).	1+1+1+1+1
34	Plant: nucleus, wall, chloroplast, large vacuole (1); Animal: nucleus, no wall, no chloroplast, small vacuoles (1); Bacteria: no true nucleus, wall present, no chloroplast (1); Bacteria are the smallest (1); Bacteria have a nucleoid (1).	1+1+1+1+1
35	Source: infected person's faeces (1); Contaminated water (1); Healthy person drinks it (1); Break: safe water, sanitation (1); Treatment with ORS and care (1).	1+1+1+1+1
36	Set-up with wire, cell, switch, compass (1); Needle deflects when current flows (1); Reversing gives opposite deflection (1); Above and below the wire deflections are opposite (1); Conclusion: magnetic field around the wire (1).	1+1+1+1+1
	OR: Readings increase with roughness (1); Friction depends on nature of surface (1); Carpet has the highest (1); Adding mass increases normal force (1); Readings increase (1).	1+1+1+1+1
37	Tractor area 1.2 square metres (1); Pressure 36000/1.2 = 30000 Pa (1); Person: 600/0.05 = 12000 Pa (1); Tractor pressure is greater but wheels are wide (1); Wide tyres prevent sinking (1).	1+1+1+1+1
	OR: Place a crystal in water (1); Colour spreads without stirring (1); Motion of particles (1); Spaces: dissolves with little volume change (1); Faster in warm water (1).	1+1+1+1+1
38	Mixture: magnet separates iron (1); Mixture retains yellow and grey colours (1); After heating: black non-magnetic iron sulphide (1); New properties (1); Compound formed by chemical change (1).	1+1+1+1+1
	OR: Variable: temperature of water (1); Same volume of water in each beaker (1); Add solute until saturated and record the mass (1); Table of masses at 20, 40, 60 degrees Celsius (1); Both rise with temperature, sugar more (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** sulphur	Sulphur is a non-metal; mercury is a metal that is liquid at room temperature.
2	**(c)** colloid	Tyndall effect.
3	**(c)** 20 cm	R = 2f.
4	**(c)** 354	12 x 29.5.
5	**(a)** oxygen	Photosynthesis.
6	**(d)** 105 L	21/100 x 500.
7	**(c)** check the experimental method	Several outliers suggest a faulty procedure.
8	**(c)** provide contrast	Stains help visibility.
9	**(a)** white blood cells and antibodies	Defensive cells and proteins.
10	**(d)** compass needle deflection	Oersted.
11	**(a)** change in colour	Force does not change colour.
12	**(d)** the same type of storm with different names	Tropical cyclones.
13	**(c)** liquid	Liquefaction.
14	**(c)** sonorous and hard	Alloy property.
15	**(b)** more concentrated	More solute.
16	**(a)** principal focus	Focus.
17	**(c)** west after sunset	Young Moon.
18	**(a)** organic waste rots in it	Decomposition uses oxygen.
19	**(a)** 3 per cent	Most is salty.
20	**(c)** length of the string	Only mass is changed.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Particles of one fit into the spaces of the other (1); This shows that there are gaps between particles (1).	1+1
22	H and O are chemically combined in a fixed ratio (1); It has properties different from H2 and O2 (1).	1+1
23	Solution = 100 g (1); Percentage = 18 per cent (1).	1+1
24	i = 60 degrees (1); Total angle = 120 degrees (1).	1+1
25	Shadow = height (1); Angle = 45 degrees (1).	1+1
26	Energy decreases at each trophic level (1); Less energy is available to support them (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Mass of solution = 175 g (1); 35/175 x 100 (1); = 20 per cent (1).	1+1+1
28	New i = 50 degrees (or 30) (1); Reflected ray turns by twice 10 (1); = 20 degrees (1).	1+1+1
29	9 am: long, west (1); 12 noon: shortest (1); 3 pm: long, east and symmetrical with 9 am (1).	1+1+1
30	Algae: producer (1); Zooplankton: primary consumer (1); Fish and heron: secondary and tertiary consumers (1).	1+1+1
31	Radius = crust + mantle + core (1); Core = 6400 - 35 - 2900 (1); = 3465 km (1).	1+1+1
	OR: Hand spans differ in size (1); Non-standard units give different values (1); Use a metre scale (standard unit) (1).	1+1+1
32	Actual = 4 / 200 = 0.02 mm (1); 1 mm = 1000 micrometres (1); 0.02 mm = 20 micrometres (1).	1+1+1
	OR: First aid is immediate help before a doctor (1); Medical treatment is by a doctor (1); e.g. cleaning and bandaging a cut (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Mirror, white paper, pins, protractor (1); Draw normal and incident ray (1); Fix pins on reflected ray (1); Measure i and r (1); i = r, all in one plane (1).	1+1+1+1+1
34	Fix a vertical stick (1); Mark the tip of the shadow every 10 minutes (1); Shortest shadow is local noon (1); It lies along the north-south line (1); Compare with clock time (1).	1+1+1+1+1
35	Grass to grasshopper to frog to snake to hawk (1); Grass to mouse to snake to hawk (1); Hawk eats mice and snakes (1); Without snakes mice and frogs increase (1); Hawks have fewer prey; grass is overgrazed (1).	1+1+1+1+1
36	Crust: 5 to 70 km, solid rock (1); Mantle: about 2900 km, hot semi-solid rock (1); Outer core: liquid iron-nickel (1); Inner core: solid iron-nickel (1); Seismic waves reveal it (1).	1+1+1+1+1
	OR: Independent: amount of fertiliser (1); Dependent: height of plants (1); Hypothesis: more fertiliser gives more height (1); Keep soil, water, light same (1); Use many plants for each amount (1).	1+1+1+1+1
37	Non-cellular, extremely small (1); Multiply inside host cells (1); Diseases: influenza, dengue, measles (1); Prevention: vaccination (1); Hygiene and avoiding crowded places (1).	1+1+1+1+1
	OR: River water may contain pathogens (1); Causes cholera, typhoid (1); Boil or filter (1); Chlorinate (1); Store in clean covered vessels (1).	1+1+1+1+1
38	Fuse and MCB: excess current (1); Earthing: leak current (1); Insulation: shocks (1); Switch on live wire (1); Proper plugs, no overloading (1).	1+1+1+1+1
	OR: Feather: large air resistance (1); Stone: small compared with weight (1); Parachutist: opens a parachute (1); Air resistance grows with speed (1); Reaches constant speed (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** a leaf	Only plant cells have all three.
2	**(b)** contaminated drinking water	Flood mixes sewage with water.
3	**(a)** nichrome	Higher resistance.
4	**(c)** 8 N	20 - 12 = 8 N.
5	**(d)** 500 Pa	20/0.04 = 500 Pa.
6	**(a)** particles of salt fit in the gaps between water particles	Spaces between particles.
7	**(c)** NH_3 and CH_4	Ammonia and methane.
8	**(c)** 30 g	0.12 x 250.
9	**(a)** 40 degrees	i + r = 80.
10	**(c)** to the south of the observer	Shadow is opposite to the Sun.
11	**(a)** foxes are removed	Fewer predators.
12	**(c)** a thick CO2 atmosphere	Runaway greenhouse.
13	**(b)** cm	Height is length.
14	**(a)** 10x	400 / 40 = 10.
15	**(b)** before exposure to disease	Preventive.
16	**(b)** the current is switched off	Electromagnet demagnetises.
17	**(b)** it induces opposite charge near it	Charge separation.
18	**(c)** outside pressure is lower than inside	Lower external pressure.
19	**(a)** boiling point (condensation point)	Condensation.
20	**(a)** copper	Copper is an element; the others are mixtures of metals.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	The pressure is enormous (1); It keeps the iron-nickel solid (1).	1+1
22	Senses can be fooled or are not exact (1); Some things are unsafe to touch or taste (1).	1+1
23	Prokaryotic: no true nucleus (bacteria) (1); Eukaryotic: true nucleus (plants, animals) (1).	1+1
24	Vaccines prevent disease; antibiotics treat bacterial infection (1); Vaccines train immunity; antibiotics kill bacteria (1).	1+1
25	More turns produce a stronger magnetic field (1); More force on iron pins (1).	1+1
26	Spikes increase friction with the ground (1); Prevents slipping (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	It switches off when current is too large (1); Using a magnet or heating strip (1); Can be reset (1).	1+1+1
28	Moon's gravity is less (1); Weight is gravitational force (1); Mass is the amount of matter (1).	1+1+1
29	Cyclone: large, spirals over warm sea (1); Tornado: small, funnel on land (1); Thunderstorm: lightning and rain (1).	1+1+1
30	Vapour: particles far apart (1); Condenses: closer, slide (1); Freezes: fixed in a pattern (1).	1+1+1
31	Lustrous (1); Malleable and ductile (1); Do not rust easily (1).	1+1+1
	OR: Prepare a hot saturated solution (1); Filter and cool slowly (1); Crystals form and are dried (1).	1+1+1
32	Wide field of view (1); Image erect (1); Objects appear smaller and farther (1).	1+1+1
	OR: Gregorian: solar, 365 or 366 days (1); Islamic: lunar, about 354 days (1); Hindu: lunisolar with extra months (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Before: warnings, shelters, supplies (1); Before: secure loose objects (1); During: stay inside (1); After: avoid flood water and wires (1); Help and report (1).	1+1+1+1+1
34	Compress the gas (1); Cool it (1); LPG (1); CNG and liquid oxygen (1); Liquid nitrogen (1).	1+1+1+1+1
35	Iron in blood (1); Calcium in bones (1); Oxygen to breathe (1); Copper wires (1); Carbon fuels (1).	1+1+1+1+1
36	Solubility decreases (1); Crystals separate (1); Remaining solution stays saturated (1); Mass of crystals depends on the difference (1); Used in crystal preparation (1).	1+1+1+1+1
	OR: Minimum height = 0.9 m (1); Top edge at half the distance of head and eyes (1); Bottom edge at half the distance of feet and eyes (1); Independent of distance from mirror (1); Mirror length is half (1).	1+1+1+1+1
37	Latitude equals the altitude, so 25 degrees North (1); Find the Big Dipper (1); Extend the pointer stars (1); Five times the gap (1); Pole Star found (1).	1+1+1+1+1
	OR: Pond: algae, fish (1); Grassland: grass, herbivores (1); Decomposers present in both (1); Abiotic differs (1); Food chain examples (1).	1+1+1+1+1
38	CFCs from refrigerators (1); Release chlorine in the stratosphere (1); UV reaches Earth (1); Montreal Protocol (1); Layer slowly recovering (1).	1+1+1+1+1
	OR: Time period rises with length (1); Rise is not proportional (1); Conclusion: longer pendulum, longer period (1); Constant: mass of bob (1); Constant: angle of swing (1).	1+1+1+1+1

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